

Recent Perspectives of Technological Advances in Sports Biomechanics

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ABSTRACT

Sports biomechanics is a fundamental sports science dedicated towards increasing performance of athletes along with prevention of risk injuries. The prime areas of interest include observation as well as analyses of fundamental movements in every sport. Following observations, key physical principles of mechanics are applied to evaluate efficacy as well as corrections in movements. The above observations help scientists to understand the relationship between kinetics and physical performance of athletes. As a result, sports biomechanics provides the safest and quickest possible way for meeting the performance goals of athletes in every sport. However, for the above goals, a sports Biomechanist requires technology support in the form of latest instrumentation, software etc. Therefore, the present review is aimed to provide information about latest research innovations in technology available in the field of sports biomechanics. The review shall definitely help athletes, scientists and related professionals working in this niche of sports science.

Key words

sports biomechanics, technology, sports science,

INTRODUCTION

Sports Biomechanics is a branch of kinesiology that studies human movements using mechanical principles (1). It mainly involves observation as well as optimization of movements in order to maximise athlete's performance and minimize his/her risks of injuries (2). The movement analyses in sports biomechanics require comprehensive in-depth observation of every movement under consideration to understand geometry of the motion

(kinematics) by recording displacement, velocity and acceleration (3). On the other hand, Biomechanist also focus on the forces responsible for the movements (kinetics) including internal forces (from muscles), external forces (friction, external load, gravity, ground reaction force) (4). Both kinematic and kinetic data of an athlete are exploited utilizing mechanical principles to elucidate complete mechanics of the movement under consideration for assessment of efficacy and risk of injury.

The technology support is essential for observation of both kinematics as well as kinetics of athlete's movements. The kinematic studies involve observation of motion without consideration of forces (5). The detailed kinematic results also includes description of joints, muscles involved in various movements. In addition, velocity, displacement, acceleration are other important observations of Kinematic analysis. So, technological tools that are utilized for this purpose are digital cameras (6). We can observe movement by our vision alone but eyes are not able to record motion. So, cameras are necessary for the recording purpose. Furthermore, many sports movements are really fast that human eye is not even able to observe all minute details. The cameras with latest technological innovations are discussed in detail in the later sections of this article. Further, some latest wearable gadgets are also now available for indirect kinematic recordings right from the field. On the other hand, kinetics studies focus on the forces like ground reaction forces, internal forces, external forces etc. The force platform is the key instrument utilized in sports biomechanics for kinetic studies involving ground reaction forces (7). Further, additional data on internal muscular forces of athlete are recorded by electromyography (EMG sensors) integrated with force platform data. Latest technological advances in kinetic studies involved use of wireless sensors for decoding this information right from

the field. All these advances in technology in kinetic studies are discussed later in detail. Also, optoelectronics is the latest and the most comprehensive tool being utilized the world over for in-depth analysis of elite athletes for performance enhancement and assessment of risk of injuries (8). It is commonly known as 3-D motion capture analysis and is discussed in detail in the concluding sections of this review. Besides this, some other latest technological advances are also discussed like instrumented treadmill, dynamometer, that are part of modern biomechanics lab. So, this review is first of kind that explored all available recent technological advances for biomechanical analyses in sports.

Technological advances in Kinematic studies (Video analysis)

The kinematic analysis focuses only on the motion without considering forces. So, it involves use of high speed cameras for recording of sports movements, technically known as, video analysis. It is broadly divided into broad categories viz. qualitative video analysis (involves analysis without quantification or measurements) and quantitative video analysis (involves analysis with measurements). Furthermore, when motion is analysed in single plane using one camera, it is called as 2D video analysis while, when more than one plane is analysed using multiple cameras it is called as 3D video analysis. Quantification is performed by digitizing movements performed by body segments into virtual horizontal

and vertical coordinates in the computer. So, cameras that are utilized in kinematic studies have high frame rates (60 fps to 1000 fps) leading to blur free super slow motion video capture of sports movements (10). The prime advantage of these cameras are that they are very simple to setup and provide sequential two-dimensional images of movement at specific time intervals depending on the speed of the camera (11). Moreover, these cameras offer excellent and economical video analysis setups for sports movements.. Also, these cameras are compatible with low cost digitising software and most of the cameras even work with open source software like kinovea. These cameras along with digitising software give information like displacement, velocity, acceleration, joint angles instantly. On the other hand, there is one essential requirement of good lighting source for proper usage of these high speed cameras. Otherwise, the clarity in the video footage will be lost. Some of the available models of high speed cameras are Phantom Miro LC320, Optronis, Sony, Simi. Selection of particular model and frame speed depends on movement being analysed. The fast movements like badminton shots definitely require high frame rates for accurate kinematic analyses.

Technological advances in Kinetic studies (Force analysis)

Biomechanics is also dedicated towards study of forces and their effects on athletes (12). The comprehensive analysis of forces help sports scientists

in understanding athlete's execution of a skill or physical development (13). For instance, force data in a vertical jump helps in understanding explosive abilities of the athlete. The gold standard instrument utilized for the measurement of ground reaction forces during sports movements is force platform (14). The ground reaction forces measured by force platform, provide scientists with complete information about the interaction of athlete with ground. This interaction data of forces lead to evaluation of many important performance parameters like take off force, landing impact, power, explosive strength, flight time, ground contact time etc. Moreover, the information available from force plate data is highly exclusive and not necessarily available from other testing methods (15). There are variety of jump tests like counter movement jump, drop jump are utilized for evaluation of various biomechanical qualities of athletes. These parameters are practically very helpful in designing training protocols, talent selection, return to play protocols in case of sports injuries.

The recent advances in the force platforms have made them fully automated with smartly designed tests protocols for fast and accurate kinetic analyses (16). Further, force platforms are broadly divided into two types on the basis of type of force transducer used viz. piezoelectric and strain gauge. The piezoelectric model has high accuracy with very low noise but come with high price tags. The piezoelectric force

platforms are more preferred choice for biomechanical research purpose. On the other hand, strain gauge force platforms are on economical side and are preferred choice of coaches. Moreover, nowadays portable force platforms are also available that allow portability for kinetic analysis even in the field. Recent advances in software associated with force platform have made kinetic analysis fast, fully automated with ability to provide real time information with ready-made performance profiles for quick comparisons (17). Technical advances in force platforms have also made it possible to choose from variety of dimensions and capacities to measure ground reaction forces on the basis of sport being analysed. For instance, ground reaction forces generated by heavy weight lifters is significantly higher in comparison to sprinters. So, ground reaction force measuring ability of force platform is different in both the sports. Hence, the availability of variety of force platform models with different capacities to measure ground reaction forces made them highly cost effective. The leading brands in force platforms include Hawkin Dynamics, Kistler, Pasco Scientific, ForceDecks.

Wearable technology for kinematic and kinetic analyses

The recent technological innovations in sports biomechanics led to the development of light weight, compact wearable sensors for kinematic and kinetic analysis of athletes right from the field (18). These wearable

sensors have made it possible to get crucial biomechanical data like acceleration, displacement, force data including ground reaction forces and internal forces from muscles, in real time right during training competitions. Coaches, scientists have access to above data on their mobile devices, tabs, laptops etc. with the help of innovative easy to use software. The above sensors are the most convenient and the only ways to get real time information about athletes from the field. This helps scientists, coaches in better understanding of sports movements during play and reveals differences (if any) in the performance of athletes in comparison to laboratory results (19). This additional information from the field significantly helps in improving athletes performance and reducing risks of injuries. Technically, these wearable sensors are known as Inertial measurement units (IMU) and micro electromechanical sensors (MEMS) which are equipped with combination of magnetometers, accelerometers and gyroscopes. Also, wearable sensors like flex sensors, such as those produced by Spectra Symbol (Salt Lake City, UT, USA), have ability to track joint motion through means of changes in resistance when a force is applied to the sensor (20).

Recently, wearable sensors have been used during football training matches for monitoring head concussions by measuring linear and angular head accelerations upon impact (21). In this way they are also useful with

regard to safety prospects of athletes while in game. Also, sensors like GPS tracking devices when placed on the upper back helps to quantify the difference in the amount of work done in different player positions as well as the game intensities (22). Some sensors are also helpful in monitoring multiple athletes in team games at once in real time, and , even allow both ways interactions between athletes and coaches too.

Further, the VERT inertial measurement unit (IMU) system (Version 2.0, Mayfonk Inc., Fort Lauderdale, FL, USA) to quantify jump height accurately (23) in the field . The VERT IMU is commercially available, as are other systems such as the KINEXON sensor (Kinexon Sports and Media GmbH, Munich, Germany).The wireless sensors also helpful in assessment of internal forces produced by muscles using wireless electromyography (EMG) sensors. Furthermore, these sensors have ability to get integrated with Force platform signals for making the study and analysis comprehensive. The only limitation with some of these wearable sensors is that information provided is not as accurate as laboratory information. The prime reasons include the presence of ferromagnetic objects can distort measurements from inertial-based systems (24). Furthermore, during wireless transfer of data from field there are chances of interference from mobile phones or other devices

leading to loss of some signals (25). Besides this, continuous efforts are being made to enhance the accuracy from these wireless sensors. The leading brands involved in the production of various wireless sensors are Xsense ,TDK, Vicon etc. Optoelectronics systems advances in sports biomechanics Optoelectronic systems are basically the advanced camera systems that convert optical light signal into electrical signal for quantification purposes. They are popularly called as motion capture systems and are considered as gold standard systems for sports motion analysis. The cameras used in optoelectronic systems are broadly divide into two types on the basis of type of body markers used. (26). The cameras that utilize active markers (markers are like LEDs producing their own light) are called active optoelectronic system cameras. On the other hand, cameras that use passive markers (markers don't produce light, instead they reflect) are known as passive optoelectronic system cameras. The correct placement of markers on the athlete body is essential for motion capture in both types of systems. However, the active marker systems are way more accurate in comparison too passive systems (27). So, sports involving complex movements require use of active marker systems but for simple sport movements passive marker systems are sufficient. Both systems involve construction of 3- dimensional model of athletes under consideration in

real time with the help of computational methods using dedicated software for model construction. The recent technological advances have made this whole process very fast, accurate and fully automated (28). Moreover, these systems nowadays provide smart sport specific profiles for fast and accurate analysis. The leading brands involved in the production of optoelectronic systems are Vicon, Optitrack, and mocap motion analysis systems. All these leading brands allow full integration of signal from motion capture system with other biomechanics instruments including force platform, inertial sensors, EMG sensors for generation of detailed comprehensive information in one go. Moreover, the digitally constructed 3D model of an athlete could also be utilized for simulations and predictions for planning of better training protocols. The coaches can clearly see the potential as well as injury risks of any new training protocol or technique in advance with the help of simulations. Further, capture volume is also important factor in selection of number of cameras in a motion capture system (29). The higher the number of cameras, more is the capture volume. For instance, a sprinter movements needs high capture volume, hence more number of cameras are essential for sprinting movements.

Recently, a third category of motion capture system has been developed, known as marker less

motion capture system (30). This system is highly flexible as it constructs 3D model without any requirement of markers on the athlete body. These systems work best for the sprinters who sometime don't want to wear body markers. Furthermore, these systems are available as combo version too allowing use of both technologies in one system. However, these systems are not accurate as marker based systems. The leading brand involved in the production of these kind of systems are Biomechanics and Captury.

Other important instruments in Biomechanics labs and technical advancements

Isokinetic dynamometer and instrumented treadmill are also important instruments of modern biomechanics laboratory (31). All these instruments have shown tremendous technical advancements in recent years. Isokinetic dynamometry is a method of quantitative myometry that uses hydraulic or electromagnetic instrument to impose constant velocity movement at pre-set angular velocities. It is employed exclusively for assessing the intensity of performance of voluntarily contracting muscles of athletes. An isokinetic training routine is helpful for athletes and others overcoming injuries, as it often prevents re-injury of the muscle being worked. It checks and confirms perfect range of motions and muscular strength post-surgery. In addition, this instrument also useful in assessment of muscle endurance as well

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